

MTH205 Analysis in one variable

Tutorial 7

DATE: **21.03.2025**

(1) Show that the function $f(x) = \sqrt[3]{x}$ is not differentiable at $x = 0$.

(2) Define $g: \mathbb{R} \rightarrow \mathbb{R}$ by

$$g(x) = \begin{cases} x^2, & \text{if } x \in \mathbb{Q}; \\ 0, & \text{if } x \in \mathbb{R} \setminus \mathbb{Q}. \end{cases}$$

Show that g is differentiable at $x = 0$ and find $g'(0)$.

(3) Prove the following statements.

(a) if an even (**odd**) function $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable in $\mathbb{R}$ then the derivative f' is an odd (**even**) function.

(b) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and suppose that for every $\varepsilon > 0$ there is a $\delta > 0$ such that

$$\left| \frac{f(a+h) - f(a)}{h} - f'(a) \right| < \varepsilon,$$

for every $a \in \mathbb{R}$ and $h \in (-\delta, \delta) \setminus \{0\}$ then f' is continuous.

(c) Suppose that $f: \mathbb{R} \rightarrow \mathbb{R}$ is differentiable at a point $c \in \mathbb{R}$ with $f(c) = 0$. Then $g(x) := |f(x)|$ is differentiable at c if and only if $f'(c) = 0$.

(4) Determine where each of the following functions from $\mathbb{R}$ to $\mathbb{R}$ is differentiable and find the derivative.

(a) $u(x) = |x| + |x + 1|$

(b) $v(x) = 2x + |x|$

(c) $r(x) = x|x|$

(d) $t(x) = |\sin(x)|$.